

Hosted Checkout

MIGRATION GUIDE



payment gateway services

Hosted Checkout



About Hosted Checkout

Our flexible Hosted Checkout solution provides you with a seamless, secure e-commerce checkout through one simple connection. We host a secure checkout experience and process transaction requests for you, so you never need to touch sensitive payment data. This allows you to focus on growing your business without the complexity of managing payments.

About this overview

This document is designed to assist merchants in understanding their current Hosted Checkout integration version and planning towards upgrading to the latest Hosted Checkout experience. If you wish to skip ahead to find out how to upgrade, see the ['Steps for Upgrading'](#).



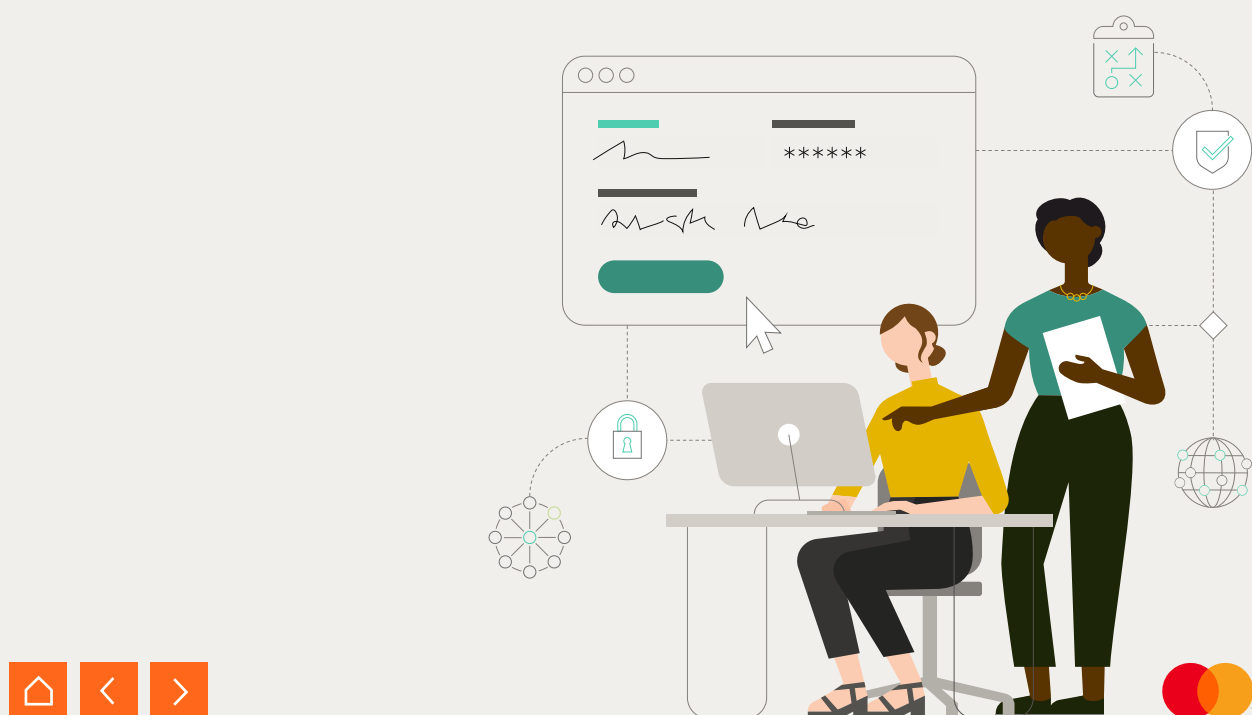
Index

1. Key features of Hosted Checkout >
2. Benefits of upgrading >
3. How does the latest Hosted Checkout differ? >
4. Important new features >
5. Steps for upgrading >
6. Steps for integrating with Click to Pay & EMV 3DS >
7. Click to Pay in Hosted Checkout >
8. Enabling EMV 3DS >
9. Testing >
10. How to tell which API version you're using >
11. Frequently asked questions >

Key features

The new Hosted Checkout experience has been fully redesigned to enhance customer experience and unlock greater choice with innovative payment solutions:

- Access our **leading global network** of over 180 acquirer connections
- Accept payment methods anywhere in the world:
 - Process card payments including Mastercard, Visa and American Express
 - Offer new, more secure ways to pay such as **Click to Pay**
 - Provide alternative payment methods including **PayPal** and **Klarna**
 - Support **domestic schemes, local payment methods** and **languages**
- The latest security and authentication features including **EMV 3DS 2.0**
- Mastercard Sonic Brand at Checkout to reinforce peace of mind at checkout
- Market leading **fraud prevention** tools including NuDetect and Brighterion
- Ongoing access to the **latest product developments** and innovation



Benefits of upgrading to the new Hosted Checkout



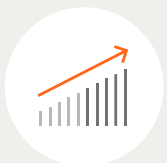
Superior customer experience by default:

- Ability to embed into your checkout or have us host on your behalf
- Fully mobile responsive across all devices, enabling a seamless customer experience
- Modular user interface to customise your checkout journey
- WCAG (Web Content Accessibility Guidelines) 2.0 Level AA compliant



Offer the payment methods consumers want:

- Offer Click to Pay, which helps consumers speed through checkout
- Toggle on/off alternative payment methods like PayPal and Klarna
- Ability to offer localized domestic payment options that your customers will love



Grow your sales:

- Our connections enable you to accept payments globally, while meeting local requirements
- Potential to improve authorization rates with EMV 3DS 2.0
- Reduce cart abandonment with a refined mobile friendly user experience
- Multilingual capabilities to help you scale globally

Benefits of upgrading to the new Hosted Checkout



Minimize your risk:

- Utilize tokenization for best-in-class secure payments
- Allow us to manage your payment page and processing to reduce PCI DSS requirements
- Leverage out-of-the-box fraud solutions such as NuData and Brighterion to reduce fraud



Remain innovative and drive conversion:

- Minimize further development costs, with ongoing access to new and innovative capabilities via your existing integration
- Offer the payment methods your customers want now, and in the future

How does the new Hosted Checkout differ compared to previous versions?

1

Full end-to-end re-design of the user experience.

2

Fully responsive design across mobile, tablet and desktop devices.

3

Inclusion of industry leading 'Click to Pay' checkout functionality.

4

The latest fraud tools including NuData and Brighterion.

5

EMV 3DS 2.0 is now supported.

6

"Lightbox mode" has been replaced with "embedded mode" for a friendlier customer experience.

7

All new and future functionality will be released on the latest Hosted Checkout.

More information about important new features

Click to Pay

- Click to Pay is an intelligent, password-free online guest checkout option that provides a quick and easy experience on any device. It provides a single button and a standardized flow for all participating card schemes.
- Built off EMVCo's secure remote commerce (SRC) specification and replacing Masterpass, Visa Checkout and Amex Express Checkout, Click to Pay offers the following benefits:
 - Intelligent consumer recognition 'embedded' into the flow for faster guest checkout across all devices (with no passwords to remember).
 - A reduction in clicks and manual card entry to increase conversion and create frictionless experiences.
 - Potential for higher authorization approval rates and reduction of preventable declines with network tokenization.
 - Secure exchange of payment data including card details, billing, and shipping address details.

EMV 3DS

- EMV 3DS 2.0, published by EMVCo, is the new benchmark for faster, frictionless transactions and reduced fraud.
- The new standard in payment transaction acceptance, it offers richer data transfer and greater flexibility for authentication.
- EMV 3DS 2.0 provides enhanced authentication making online payments more secure and reducing fraudulent transactions for the benefit of merchants, issuers, and cardholders.

Steps for upgrading

Step 1

Add or update server-to-server call

In your current implementation of Hosted Checkout, you may or may not have used an authenticated session.

Using an authenticated session increases the security of Hosted Checkout by:

- Requiring a server-to-server call to create a secure session
- Ensuring that the session passed in via the JavaScript call matches that created by the server

Previous versions of Hosted Checkout did not require this call. However, for security reasons, it is now required for modern checkout

Follow these steps if you are already making a server-to-server call to create an authenticated session

If you're already creating a session using the [CREATE_CHECKOUT_SESSION](#) endpoint, all you have to do is update the apiOperation value in the request body:

Before	After
<pre>{ "apiOperation": "CREATE_CHECKOUT_SESSION" }</pre>	<pre>{ "apiOperation": "INITIATE_CHECKOUT" }</pre>

There may be request fields that you are currently passing that have since been deprecated, so please consult the current [INITIATE_CHECKOUT](#) documentation.

Follow these steps if you are **NOT** currently making a server-to-server call to create an authenticated session

If you are not currently doing so, you will have to do this to upgrade to modern Hosted Checkout. [This document](#) will guide you on how to set up this request.

Step 2

Update the JavaScript tag source to the new URL

You will use a new JavaScript source URL when upgrading to modern Hosted Checkout:

Before	After
<code><script src="https://ap-gateway.mastercard.com/checkout/version/57/checkout.js"></script></code>	<code><script src="https://ap-gateway.mastercard.com/static/checkout/checkout.min.js"></script></code>

Step 3

Ensure you're passing session ID in your Checkout.configure() call

You'll need to pass the authenticated session ID into your Checkout.configure() call:

```
Checkout.configure({  
  session: {  
    id: '<your_initiate_checkout_session_ID>  
  }  
});
```

See also [documentation for configure\(\)](#).

Step 4

Replace any usage of Checkout.showLightbox() with Checkout.showEmbeddedPage()

The showLightbox() method has been deprecated in favor of showEmbeddedPage(). See also [documentation for showEmbeddedPage\(\)](#).

Before	After
<pre>Checkout.showLightbox()</pre>	<pre>Checkout.showEmbeddedPage('#embedded-host')</pre> or you can showEmbeddedPage in a lightbox: <pre>Checkout.showEmbeddedPage('#lightbox-host', () => { \$('#embedded-lightbox').modal() } // tell us how to launch it for later use)</pre>

Steps for integrating with Click to Pay & EMV 3DS

Click to Pay in Hosted Checkout

If you are using the gateway's payment page (Hosted Checkout), Click to Pay will automatically be offered to your payers if activated in your merchant profile, and you are using API version 63 or higher when submitting the Initiate Checkout request.

While your payer's Click to Pay profile can contain cards for any supported scheme, they can only use Click to Pay for a given transaction where:

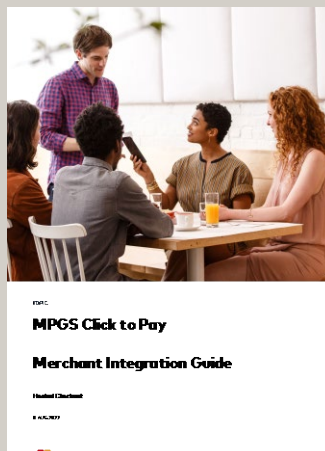
- the card scheme has been activated for SRC on your merchant profile, and
- your merchant profile is configured to process cards with this scheme and the transaction currency.

Shipping Address: Payers will not be able to select a shipping address during the Click to Pay interaction. If shipping address is required, the merchant must prompt for this prior to launching Hosted Checkout.

Billing Address: A billing address will always be collected during the Click to Pay interaction.

3-D Secure Authentication: If you are configured for 3-D Secure (3DS) authentication, Hosted Checkout will automatically perform the 3DS authentication after the Click to Pay interaction.

To review the full Click to Pay integration guide click here:



Enabling EMV 3DS

Prerequisites for all Acquirers

1. Enroll in the card scheme programs for EMV 3DS (e.g., Mastercard Identity Check, Visa Secure).
2. Add their Merchants to the scheme(s):
 - o Mastercard via Identity Solutions Services Management tool (ISSM) in Mastercard Connect, enrolment is required for all Merchants even if they used SecureCode previously.
 - o Enrolment is not required for existing Verified by Visa and American Express SafeKey Merchants. New Merchants will need to be enrolled.
3. Update their acquiring platform:
 - o Mastercard requires extra values to be passed in authorization and clearing (i.e., Enhanced AAV).

Prerequisites for Merchants *(India and Bangladesh only)*

1. The Merchant's acquirer needs to complete their Acquirer's prerequisites.
2. Additional configuration is required on the Merchant's profile in the payment gateway, for each scheme their acquirer supports.
3. Mastercard, Visa and Amex require an additional Merchant privilege. This can be configured in the Merchant Profile Page > Payment Details Page > Card Holder Verification section within the Merchant Manager Portal.

Merchant integration:

1. Integrate against Web Services API (WS-API) v57 and above
2. Follow the described steps on the help page below:
<https://ap-gateway.mastercard.com/>
3. Enable EMV 3DS via Merchant Administrator by navigating to: Admin > Integration Settings > Hosted Checkout

Testing



Testing

You can test your implementation using these steps

Ensure that you are able to successfully create a session using the [INITIATE CHECKOUT](#) API endpoint. To do this you will need your merchant ID and API password. See the [Request a Hosted Checkout Interaction](#) section of the integration guidelines. A successful response looks like this:

```
{
  "merchant": "[YOUR_MERCHANT_ID]",
  "result": "SUCCESS",
  "session": {
    "id": "SESSION0002981212879J67925298E6",
    "updateStatus": "SUCCESS",
    "version": "267f13eb01"
  },
  "successIndicator": "c05fd12a87e4412c"
}
```

Use the session ID from the successful INITIATE_CHECKOUT response to call Checkout.configure() (see step 3 above). Once you invoke Checkout.showEmbeddedPage() or Checkout.showPaymentPage() the Hosted Checkout experience will be displayed.

You can test an end-to-end transaction using the test cards listed [here](#).

How to tell which API version you're currently using

Should it be useful, you can determine the API version you are currently integrated with by looking at the request to identify the version number.

You can tell this by looking at either:

- The Javascript source URL (if using legacy checkout)
 - The legacy checkout experience includes the API version in the script URL, e.g. <https://ap-gateway.mastercard.com/checkout/version/57/checkout.js>. For modern checkout, the version number is not included in the script URL.
- Your API request URL
 - The API version will always be included in any API request URL. It is the numeric value after "/version/", e.g. <https://ap-gateway.mastercard.com/api/rest/version/67/merchant/{merchantId}/session>



Frequently asked questions

Frequently asked questions

Is Click to Pay supported by Hosted Checkout?

Yes. Click to Pay capability is an integral part of the Hosted Checkout solution. For further information relating to Click to Pay integration, see [page 15](#).

Can we customize Hosted Checkout?

Yes, with full page redirect mode you have the flexibility to add your logo and change accent color.

Is Branding Development Kit still supported?

Branding Development Kit has been removed. However, with full page redirect you have the flexibility to add your logo and change accent color.

Is SVG (Scalable Vector Graphics) format supported for logo?

Yes, SVG along with JPEG and PNG are supported formats for your logo.

Is the lightbox going away?

The lightbox is going away in v63 of Hosted Checkout; however, you will be able to customize and create a lightbox version at your end.

Is there an order summary detail in the full-page redirect and embedded options?

For full page redirect there is an order summary section that highlights the tax, shipping, etc. A merchant using the embedded option must host their own order summary page on their website.

Are there any potential impacts for customers?

Existing Hosted Checkout merchants would need to upgrade to v63 or higher of Hosted Checkout to leverage the new capabilities.

